

Anthony Hana

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EDUCATION

McMaster University

Sept 2022 - Present (Exp. Apr 2027)

Honours Mathematics and Computer Science (B.Sc.)

Hamilton, ON

- **Extra Curriculars:** McMaster Start Coding, McMaster Game Dev Club, Volleyball / Soccer Intramurals
- **Courses:** Software Development, Data Structures & Algorithms, Natural Language Processing, Numerical Linear Algebra, Mathematical Modeling, Databases, Computer Architecture

EXPERIENCE

AI & ML Engineer

September 2025 - Present

LTV.ai

Toronto, ON

- Shipped high-throughput email systems, scaling capacity by **400%** with async processing, batching, and workers.
- Optimized real-time analytic performance driving **client acquisition and retention** across high-volume datasets
- Refined LLM email generation I/O, achieving **98%** accuracy while reducing costs and token usage by **30%**
- Automated ephemeral dev environments via GitHub OIDC, unblocking parallel feature development

Software Architect

June 2025 - Present

STaBL Foundation (McMaster Start Coding)

Hamilton, ON

- Developing the Stabl.Rocks IDE, supporting workshops and university coursework for **80,000+** students
- Shipped custom Haskell plugins and functionality powering MSC's multisite platform and core course workflows
- Reduced latency and improved responsiveness by **10%** across the McMaster Start Coding platform

PROJECTS

LavaLock - DeltaHacks XII | 📱 | 🔒 | Swift, Go, C++, AWS, PostgreSQL

- Achieved **1st place** in the Swift Student Challenge at DeltaHacks XII by developing a hardware-backed password vault leveraging **physical entropy** and **GenAI** for secure credential generation
- Architected a scalable **Go/C++** backend processing hardware streams via **AWS S3** and **EC2**, enabling secure image pipelines and high-performance database transactions
- Built a native SwiftUI iOS app featuring fluid animations and secure keychain integration to generate and store cryptographically secure passwords

Posture-bot - Hack The North 2025 | 📱 | 🔒 | Python, JavaScript, Typescript, Node.js, DynamoDB

- Built an AI-powered **posture tracking system** in under 36 hours using **OpenCV** and **MediaPipe Pose** to analyze skeletal geometry and detect slouching in real time
- Implemented an **AI-generated posture report** summarizing full-session analytics, angle distributions, stability metrics, and improvement insights to personalize feedback for users

HEALense AI - SpurHacks 2025 | 📱 | 🔒 | Cloudflare, Gemini API, Typescript, React, Javascript, Tailwind CSS

- **Multimodal** fullstack app: upload a photo, query **Gemma 3**, receive answers highlighted directly on the image/chat
- Built a TS Cloudflare Worker backend that routes chat, caches in **KV**, stores images in **R2**, logs history in **D1**
- Streams user **image + prompt** payloads to the **Gemma 3** AI via gemini api and relays structured responses to clients

MacPad - Native MacOS Text Editor | 📱 | Swift, UI/UX, Gemini API

- Built **MacPad**, a native macOS Swift text editor with inline AI-powered writing suggestions
- Optimized editor performance and implemented robust multi-language text encoding for fast, reliable large-file editing

TECHNICAL SKILLS

Programming: Typescript, Javascript, Python, Go, Swift, C++, R, SQL, Haskell

Frameworks & Tech: Angular, NodeJS, React, Tailwind, FastAPI, Terraform, Handlebars, LLMOps, NextJS

Tools and Platforms: AWS (ECS, EC2, CW, S3), Git/Github (OIDC), Postman, Docker, Overleaf, Jira

Data: PostgreSQL, DynamoDB, Redis, Clickhouse, Kafka, DBeaver